SOFIA MUÑOZ RUIZ

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EDUCATION

MSc	*MSc Student in Engineering, specializing in Neuroengineering - Faculty of Engineering, University of Antioquia, Colombia*	*2025-2026*
BSc	*Engineering Physics, Department of Physics, University of Cauca, Colombia*	*2017-2024*

PROFILE

I am a highly skilled professional with a BSc in Engineering Physics and MSc student in Engineering, specializing in Neuroengineering, with a strong interdisciplinary background in biomedical signal processing, artificial intelligence, and computational neuroscience. My academic training has provided me with solid foundations in mathematics, physics, statistics, signal processing, image analysis, and data science using Python and MATLAB.
During my undergraduate studies, I worked with electroencephalographic (EEG) signals from pediatric epilepsy patients, which allowed me to develop a deeper understanding of brain activity and electrophysiological signal analysis. In collaboration with experts in biosignals, medicine, and artificial intelligence, I developed my undergraduate thesis entitled "Implementation of Artificial Intelligence Algorithms for the Analysis of Brain–Heart Relationship During Anesthesia Using Physiological Signals." This research focused on studying the interaction between the brain and heart through nonlinear analysis methods and machine learning techniques applied to EEG and electrocardiographic (ECG) signals. As part of this work, I extracted and processed data from 605 patients from the VitalDB database and applied the CRISP-DM methodology for data analysis and modeling. Additionally, I classified four anesthesia states using various machine learning models and evaluated their performance. This experience strengthened my interest in complex physiological systems and the application of AI methods in healthcare and neuroscience research.

Currently, during my master's studies, I teach laboratory courses in biosignals and systems, where I supervise student projects focused on Brain–Computer Interfaces (BCI), EEG signal processing, machine learning and deep learning. Additionally, I teach topics in Artificial Intelligence, Machine Learning, and Deep Learning applied to *biomedical engineering. These experiences have strengthened my communication and mentoring skills, enhanced my ability to explain complex concepts effectively, and deepened my expertise in these areas. In parallel, I am conducting research in collaboration with the Neuroscience Group of Antioquia and the University of Rome, focusing on the identification of neurophysiological patterns associated with the preclinical stages of Alzheimer's disease. This work involves the analysis of EEG signals recorded during interactions with serious games and aims to uncover biomarkers that may contribute to the early detection of neurodegenerative processes. This experience has allowed me to gain expertise in EEG acquisition from individuals carrying the E280A genetic mutation associated with early-onset Alzheimer's disease. As part of this work, I developed my research project entitled "Identification of Changes in Neurophysiological Patterns Associated with Preclinical Stages of Alzheimer's Disease Using Deep Learning Models and Electroencephalographic Signals Recorded During the Execution of Serious Games."*

Through this research, I have strengthened my skills in low-density EEG acquisition, neurocognitive processes signal processing, deep learning models and serious games. My experience in data analysis within healthcare and neuroscience, combined with strong analytical, research, and collaborative skills, has enabled me to work effectively in multidisciplinary research environments.

I am particularly motivated to continue exploring diverse areas of neuroscience and biomedical engineering, applying advanced computational methods to better understand brain function. My long-term goal is to contribute to the development of innovative tools and knowledge that advance our understanding of the human brain.

RESEARCH EXPERIENCE

Researcher in Neuroengineering and Biomedical Signal Processing *University of Antioquia, Medellín, Colombia* *Conducting research on the identification of neurophysiological biomarkers associated with preclinical stages of Alzheimer's disease using EEG signals and Deep Learning models.* *Acquiring and analyzing low-density EEG recordings from carriers and non-carriers of the E280A mutation associated with early-onset Alzheimer's disease.* *Working with EEG signals recorded during serious game execution tasks using the SmartMe&You platform developed in collaboration with the University of Rome.* *Applying Deep Learning and Machine Learning techniques for neurophysiological pattern recognition and cognitive state analysis.*	*2025 – Present*
Undergraduate Research Thesis — Brain–Heart Interaction During Anesthesia *University of Cauca, — Colombia*	*2023-2024*

Developed an AI-based framework for analyzing brain–heart interactions during anesthesia using physiological signals.
Processed physiological data from 605 patients obtained from the VitalDB database following the CRISP-DM methodology.
Applied nonlinear analysis techniques to study dependency and directionality between EEG and ECG signals.
Classified four anesthesia states using Machine Learning models including SVM, XGBoost, Random Forest, and Neural Networks.
Collaborated with multidisciplinary researchers in biosignals, medicine, and artificial intelligence.

EEG and Neuroimaging Analysis in Pediatric Epilepsy — *2022*

University of Cauca, Colombia
2022 – 2023

Worked with EEG signals and neuroimaging data from pediatric epilepsy patients.
Performed signal preprocessing, analysis, and interpretation of different brain wave patterns.

QUALITIES RELATED TO ACADEMIC EDUCATION

University of Antioquia *Laboratory Instructor — Biosignals and Systems* *Teaching laboratory sessions on biomedical signal acquisition and processing using Python.*	*2025 – Present*
University of Antioquia *Supervising student projects focused on Brain–Computer Interfaces (BCI), EEG processing, and signal classification.*	*2025 – Present*
University of Antioquia *Introducing students to AI, Machine Learning, and Deep Learning techniques applied to biomedical engineering and biosignal analysis.*	*2025 – Present*

STUDY VISIT ABROAD - INTERNATIONAL RESEARCH NETWORK

Research Presentation — Brain–Heart Interaction During Anesthesia Using Physiological Signals — *2024*

Center for the Heart and Brain, Gottingen, German

Presented research on the analysis of brain–heart interactions during *anesthesia using artificial intelligence and physiological signals. Discussed nonlinear analysis methods and machine learning approaches applied to EEG and ECG signals. Interacted with researchers working in biosignals, neuroscience, and cardiovascular physiology.*

Research Presentation — Brain–Heart Interaction During Anesthesia Using Physiological Signals *Faculty of Applied Computer Science, University of Augsburg* *Presented research on the analysis of brain–heart interactions during anesthesia using artificial intelligence and physiological signals. Discussed nonlinear analysis methods and machine learning approaches applied to EEG and ECG signals. Interacted with researchers working in biosignals, neuroscience, and cardiovascular physiology.*	*2024*

PUBLICATIONS

Journal Publications

Muñoz-Ruiz S, Saavedra-Ordoñez J, Valencia-Fajardo N, Spicher N, Vargas-Cañas R, Idrobo-Ávila E. Analysis of electrocardiographic signals to assess heart rate variability under surgical context: a methodological proposal. Workshop Biosignals; 2024-02-28 - 2024-03-01; Göttingen

M. S. Muñoz, C. E. S. Torres, D. M. López, R. Salazar-Cabrera, y R. Vargas-Cañas, «Automatic Detection of Epileptic Waves in Electroencephalograms Using Bag of Visual Words and Machine Learning», en Brain Informatics, vol. 12241, M. Mahmud, S. Vassanelli, M. S. Kaiser, y N. Zhong, Eds., en Lecture Notes in Computer Science, vol. 12241. , Cham: Springer International Publishing, 2020, pp. 163-172. doi: 10.1007/978-3-030-59277-6_15.

M. S. Muñoz, C. E. S. Torres, R. Salazar-Cabrera, D. M. López, y R. Vargas-Cañas, «Digital Transformation in Epilepsy Diagnosis Using Raw Images and Transfer Learning in Electroencephalograms», Sustainability, vol. 14, n.o 18, p. 11420, sep. 2022, doi: 10.3390/su141811420.

SCIENCE COMMUNICATION: PRESENTATIONS

Biosignals Workshop organized by the Biosignal Processing Group of the University Medical Center Göttingen (UMG), Germany.	*2024*
Oral Presentation: Brain–Heart Interaction During Anesthesia Using Physiological Signals *Center for the Heart and Brain, Gottingen*	*2024*
Oral Presentation: Brain–Heart Interaction During Anesthesia Using Physiological Signals	*2024*

Faculty of Applied Computer Science,University of Augsburg	
Poster Presentation: "EEG-Based Characterization of Neurophysiological Alterations in Mild Cognitive Impairment During Serious Game Interaction: Preliminary Results" *Computational Approaches to Alzheimer's Disease Symposium, Medellín, Antioquia*	*2026*

JOURNAL PUBLICATIONS (UNDER REVIEW)

Lightweight R-peak detector grounded on a synthetic QRS-template for resource-constrained devices
Corresponding Author: Doctor Ennio Idrobo
Co-Authors: Sofia Muñoz-Ruiz, B.Sc. (Eng.); Juandiego Saavedra-Ordoñez, B.Sc. (Eng.); Natalia Valencia-Fajardo, B.Sc. (Eng.); Rubiel Vargas-Cañas, Ph.D.

SCIENCE COMMUNICATION: WORKSHOP COLLABORATION

Workshop; Physiological control of effort with music. University of Ghent – University of Cauca. 2015.

AWARDS & RECOGNITIONS

University of Cauca
Honorable Mention for Undergraduate Thesis "Implementation of Artificial Intelligence Algorithms for the Analysis of Brain–Heart Relationship During Anesthesia Using Physiological Signals" Thesis supervised by Rubiel Vargas Cañas, PhD, Department of Physics, and co-supervised by Ennio Idrobo Ávila, PhD, Faculty of Applied Computer Science, University of Augsburg.

University of Antioquia
Honorable Mention for Research Poster Presentation "EEG-Based Characterization of Neurophysiological Alterations in Mild Cognitive Impairment During Serious Game Interaction: Preliminary Results", presented at the Computational Approaches to Alzheimer's Disease Symposium, an international symposium focused on computational methodologies for Alzheimer's disease research.

DAAD Study Visit Scholarship Recipient: Selected participant in the "Study Visits for Groups of Foreign Students in Germany" program awarded by the German Academic Exchange Service (DAAD).

Recognition for Project Management and Proposal Writing: Acknowledged for coordinating and contributing to the proposal development that enabled multiple students to obtain DAAD academic mobility benefits.

SCIENTIFIC GRANTS

University of Antioquia

Student Instructor Scholarship / Academic Merit Recognition

LANGUAGES

English*: Upper-Intermediate (B2.2) written and spoken communication skills, with experience in scientific writing, including research reports, academic theses, and peer-reviewed publications.*
French: *A2 (Elementary)*
Spanish*: Native language*

COMPUTER SKILLS

General-purpose programming language*: Matlab, Python, R Studio*
Electronic prototyping: *Arduino, OpenBCI*
Office skills: *MS Office, Zotero, Mendeley, creating documents, preparing documents for print, proofreading and editing copy, creating slideshows, presentation design, templates, custom slides, creating graphs and charts, presentations troubleshooting.*

PERSONAL SKILLS

Pro-active personality, detail-focused, decisive, imaginative, approachable and supportive.
Solid work ethic.
Ability to learn quickly, to adapt to changing requirements, think creatively, and to work independently and as part of a team.
Verbal and written communication. Organizational skills

References
- Dr. Ennio Idrobo-Ávila
Prof. Faculty of Applied Computer Science, University of Augsburg.
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- Rubiel Vargas-Cañas PhD.
Associate professor
SIDICO Research Group
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- Jhon Ochoa-Gomez PhD.
Associate professor
GRUNECO Research Group
Prof. Bioengineering Program. Universidad de Antioquia, Colombia
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M. Sofia Muñoz Ruiz
Sofia Muñoz Ruiz BSc